

LAB 2

Bandari Bharathkumar Reddy

University of Maryland – Baltimore County

Presented to: **Gina Scaldaferri**

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Introduction

This lab exercise focuses on the procedures used by forensic investigators to acquire digital evidence and transfer data to external storage devices without compromising its integrity. In this lab, ProDiscover and FTK Imager were used to create forensic images of source media and store them on a hard disk. Both tools generate exact images of the source drive and produce cryptographic hash values using MD5 and SHA-1 algorithms to ensure data integrity and authenticity.

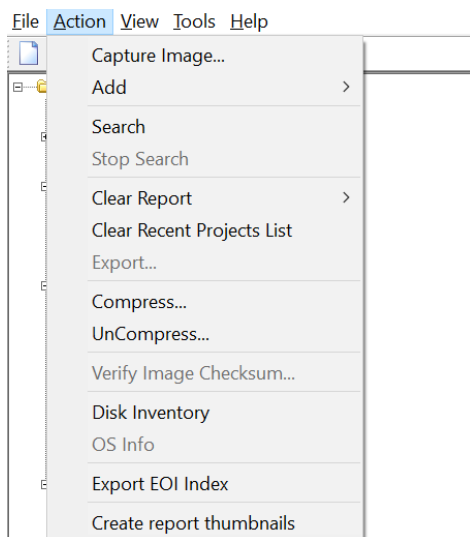
Analysis

Using Pro Discover App:

In this section, the source drive was a USB drive (U:), and the destination was C:\Users\student\Desktop\Work (although, in practice, this should be an external hard disk).

Steps To Follow:

- Launch the ProDiscover application and enter the project name and project number.
- From the top-left menu, select Action, then click Capture Image.



- In the Capture Image dialog box, select the USB drive (U:) as the source drive.
- Click the double-arrow button next to the destination field and choose C:\Users\student\Desktop\Work. Enter **USB-Pro-U** as the image file name.
- Enter the technician's name and image information, then click **OK** to begin the imaging process. (**Technician Name:** Bharath Reddy, **Image Field:** USB-Pro-U)

Capture Image ✕

Source Drive: U:\ [USB Drive] 3.997 GB ▼

Destination: C:\Users\student\Desktop\Worl >> Split

Image Format: ProDiscover Format (recommended) ▼

Total sectors to capture : 8382463 HPA

Shadow Volume Name: ▼

ProDiscover Image

Technician Name: Bharath Reddy

Image Number: USB-Pro

Description :

Compression

☐ Yes

☒ No

Password...

OK
Cancel

- Technician name: Bharath Reddy
- Image Fields: USB-Pro-U.

1.What is the resulting file format of the image?

- The .eve is the resulting file format.
- eve (Embedded Vector Editor Diagram files). Embedded Vector Editor Diagram is defined by Barry Kauler. EVE files are supported by software applications available for devices running Windows.

2.What is the hash of the image?

- MD5 - 79F0B5D05C44189598FA9611FE504AEC
- SHA1 - AFD20A96C5EEDD04AEFD649A9D381172CDD5DAF4

```
Algorithm : MD5
Hash      : 79F0B5D05C44189598FA9611FE504AEC
Path      : C:\Users\student\Desktop\work\USB-Pro-U.eve
```

```
PS C:\Users\student\Desktop\work> _
```

 Windows PowerShell

```
Algorithm : SHA1
Hash      : AFD20A96C5EEDD04AEFD649A9D381172CDD5DAF4
Path      : C:\Users\student\Desktop\work\USB-Pro-U.eve
```

```
PS C:\Users\student\Desktop\work>
```

MD5 – The MD5 (Message-digest Algorithm) hashing algorithm is a one-way cryptographic function that accepts a message of any length as input and returns as output a fixed-length digest value to be used for authenticating the original message (Mary E. Shacklett). MD5 has been deprecated for uses other than as a noncryptographic checksum to verify data integrity and detect unintentional data corruption.

What is Cryptography?

Cryptography is the study of secure communication techniques that allows only users and intended recipients of a message to view its content.

SHA1 – Secure Hash Algorithm. It works as similar as MD5. The hash value is longer than MD5. However, the speed of MD5 is fast in comparison to SHA1. SHA1 provides more secure.

The source drive is the USB drive (F:) and the destination is the C:\Users\student\Desktop\Work (Actually it must be our hard disk).

Steps To Follow:

- You can follow the same steps that you did for USB-Pro-F. But, In the source field you need to choose Drive F:
- Technician name: Bharath Reddy
- Image Field: USB-Pro-F

1.What is the resulting file format of the image?

- The .eve is the resulting file format.

2.What is the hash of the image?

- MD5 - BBCAE44CD0834DE08271CFC5CE71BC5F
- SHA1 - 8F97BA5CC91CE7C58957456DFAD016F71D4F5603

```
Algorithm : MD5
Hash      : BBCAE44CD0834DE08271CFC5CE71BC5F
Path      : C:\Users\student\Desktop\work\USB-Pro-F.eve
```

```
PS C:\Users\student\Desktop\work>
```

```
Algorithm : SHA1
Hash      : 8F97BA5CC91CE7C58957456DFAD016F71D4F5603
Path      : C:\Users\student\Desktop\work\USB-Pro-F.eve
```

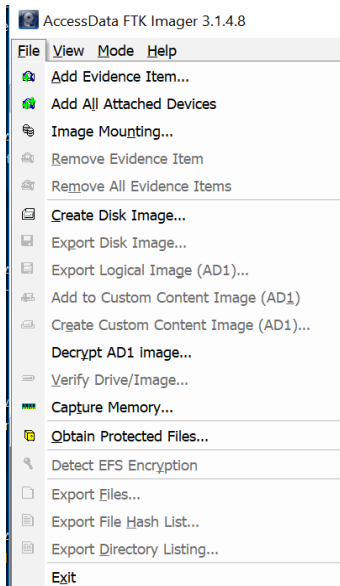
```
PS C:\Users\student\Desktop\work> _
```

Using FTK Imager App:

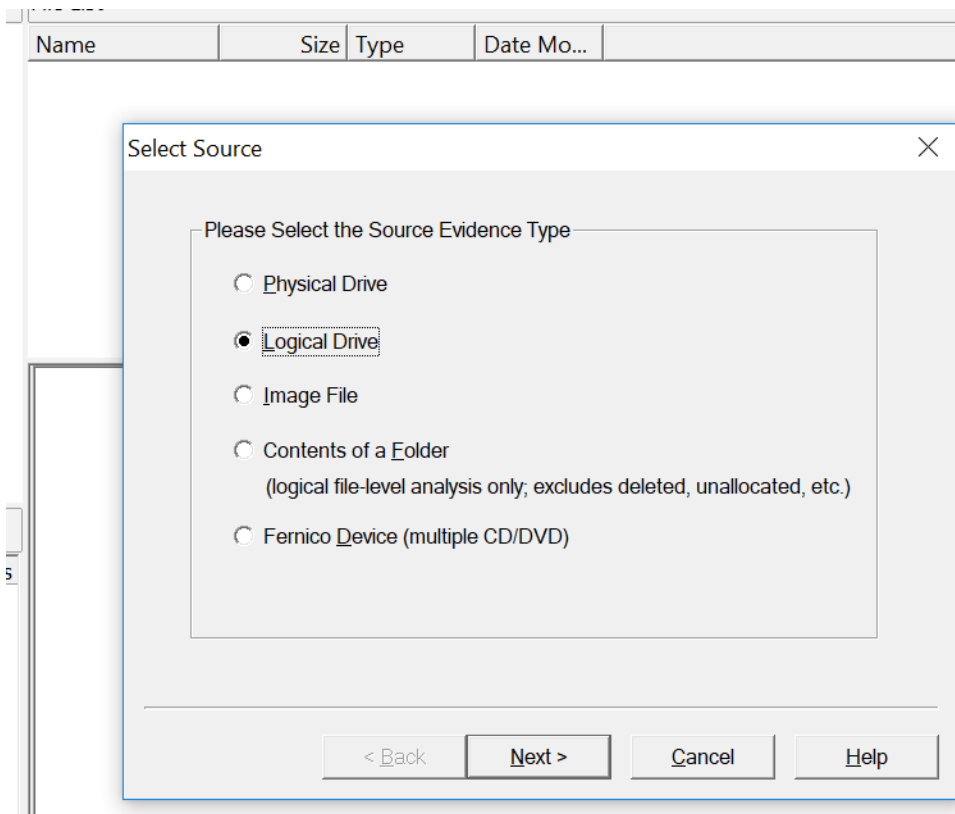
For FTK Imager, the source evidence type was set to a logical drive, and the USB drive (U:) was selected. The image type used was E01.

Steps To Follow:

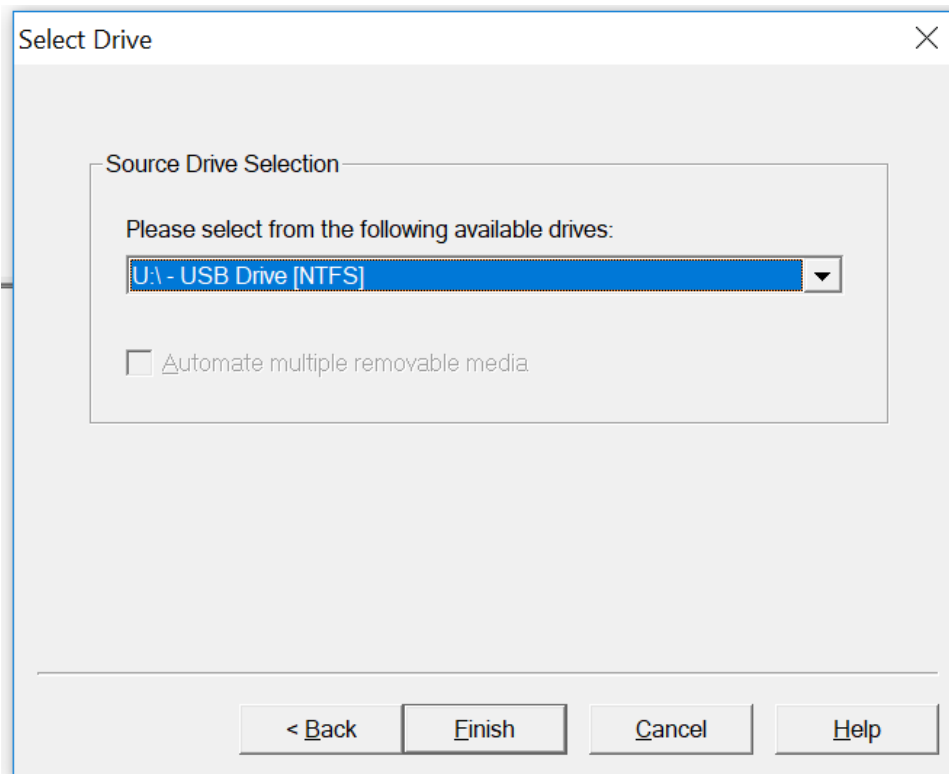
- Open FTK Imager and select **Create Disk Image** from the File menu.



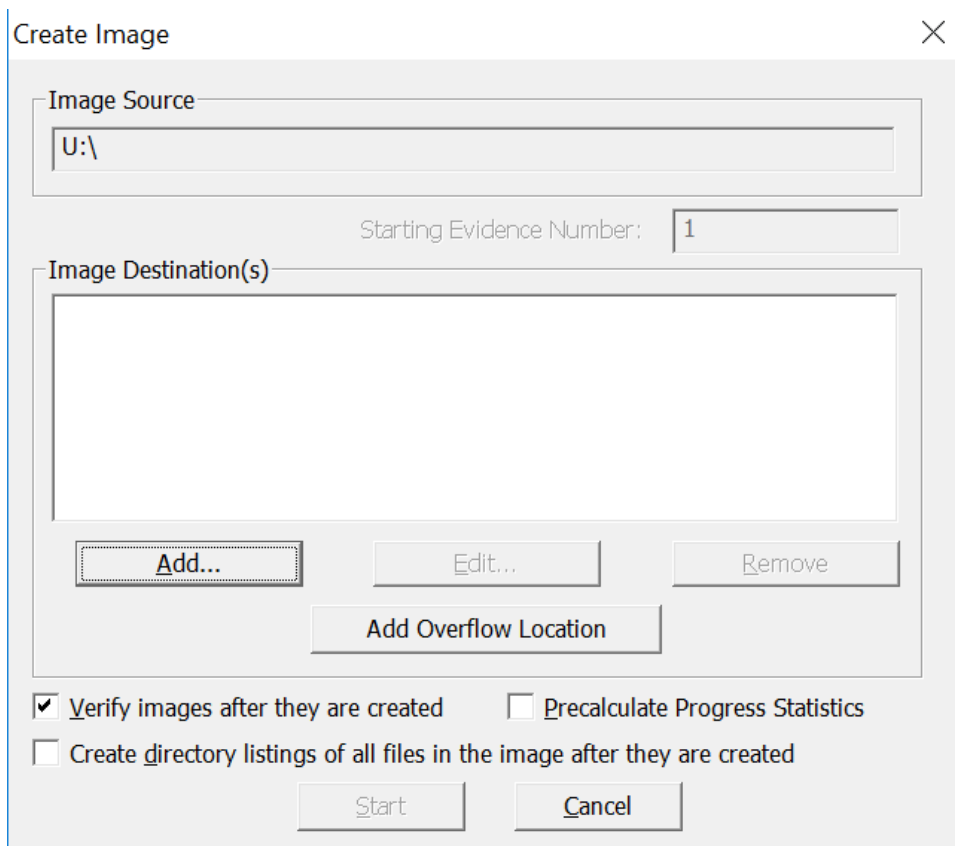
- In the source dialog, select logical drive as a Source evidence type and click Next.



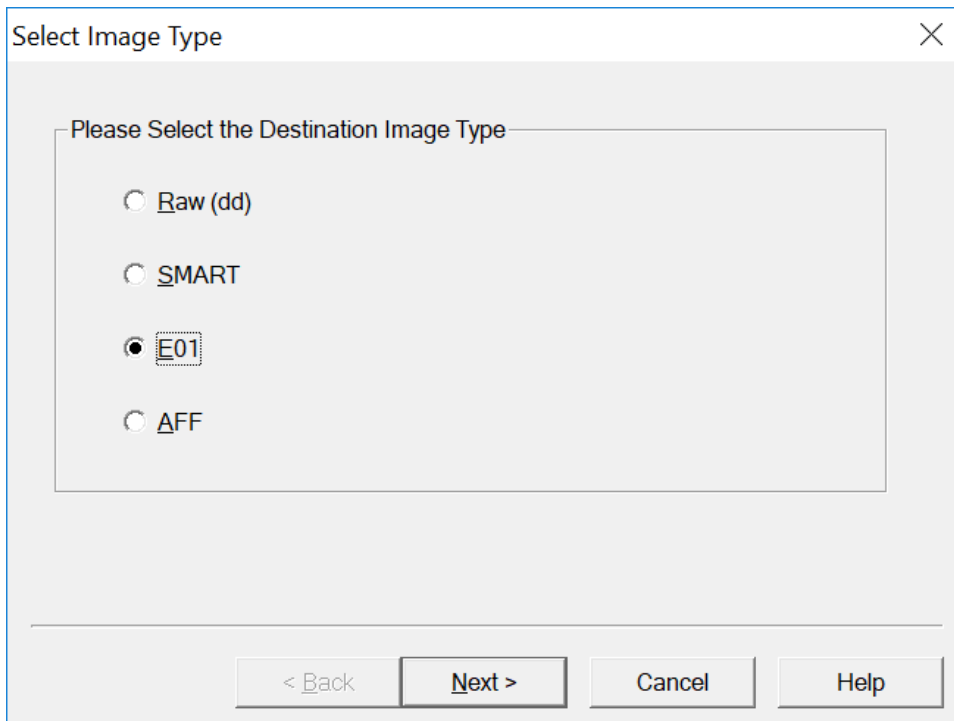
- In the select drive dialog select source drive as a USB Drive (U:), then click finish.



- In the create image dialog click on add.



- In the select Image type dialog, select E01 and click next.



- In the evidence type information dialog:
 - a. Type Examiner name in Examiner Field
 - b. Type case number as USBFTK-U
 - c. Under Unique Description mention a comment referencing this evidence potentially
 - d. deleted evidence and click NEXT to Continue.

Evidence Item Information ✕

Case Number:	USB-FTK-U
Evidence Number:	USB-FTK-U
Unique Description:	This evidence contains potentially
Examiner:	Bharath Reddy
Notes:	

< Back Next > Cancel Help

- The destination where the image is created is C:\Users\student\Desktop\Work and give file type as USB-FTK and click Finish.

Select Image Destination ✕

Image Destination Folder
C:\Users\student\Desktop\Work Browse

Image Filename (Excluding Extension)
USB-FTK

Image Fragment Size (MB) 1500
For Raw, E01, and AFF formats: 0 = do not fragment

Compression (0=None, 1=Fastest, ..., 9=Smallest) 6

Use AD Encryption ☐

< Back Finish Cancel Help

Examiner: Bharath Reddy

Case and Evidence Number: USB-FTK-U

The destination where the image is created is C:\Users\student\Desktop\Work (Actually it has to be our hard disk).

1. What is the resulting File format of the image?

- The resulting file format is EO1.

What is EO1?

EO1 (EnCase Image File Format) file storage obtains backup copies of various digital evidence, including disk images, logical file storage, etc. When an investigator (or forensic expert) uses EnCase to back up the data available on the hard drive, a physical bitstream of the data is generated.

The basic theory behind the relationship between EnCase and EO1 image file format is that when encase creates an image based on the data available on the hard disk, it divides the entire data into 640 MB blocks. Due to this data division in the 640 MB pause, multiple data files are created, by storing crucial information from the hard disk. The most particular feature of these files is that the file name remains the same (as appointed by the user), while the file extension changes.

2. What is the hash of the image?

- MD5 - 1af419c7daa1bb149efcd5cdcdb05339
- SHA1 - 72c7768243308f4a83caf1031fe91f70c2eba074

Drive/Image Verify Results	
Name	USB-FTK.E01
Sector count	8382464
MD5 Hash	
Computed hash	1af419c7daa1bb149efcd5cdcdb05339
Stored verification hash	1af419c7daa1bb149efcd5cdcdb05339
Report Hash	1af419c7daa1bb149efcd5cdcdb05339
Verify result	Match
SHA1 Hash	
Computed hash	72c7768243308f4a83caf1031fe91f70c2eb
Stored verification hash	72c7768243308f4a83caf1031fe91f70c2eb
Report Hash	72c7768243308f4a83caf1031fe91f70c2eb
Verify result	Match

Computed hash:

The hash computed during the verify process.

Stored verification hash:

The hash computed during image acquisition and stored inside the image.

Report hash:

The hash computed during image acquisition and stored in the image summary report.

It will verify all these hashes and gives us a verified result as “Match”.

Report Hash	72c7768243308f4a83caf1031fe91f70c2eb
Verify result	Match
☐ Bad Sector List	
Bad sector(s)	No bad sectors found
Close	

The source dialog is logical drive and the drive dialog is USB drive (F:) and add image type as a E01.

Steps To Follow:

- Follow the same steps as I did in the part 1. Make source drive as a USB Drive (F:) in the place of USB Drive (U:)
- Examiner: Bharath Reddy
- Case and Evidence Number: USB-FTK-F

1.What is the resulting File format of the image?

- The resulting file format is EO1

2. What is the hash of the image?

- MD5 - cf302355dbff780e924dc247cc63e6da
- SHA1 - 94cbd4f7d63d0bde686ba2f3b4d8ed4eabee55fd

Name		USB-FTK-F.E01
Sector count		8382464
MD5 Hash		
Computed hash		cf302355dbff780e924dc247cc63e6da
Stored verification hash		cf302355dbff780e924dc247cc63e6da
Report Hash		cf302355dbff780e924dc247cc63e6da
Verify result		Match
SHA1 Hash		
Computed hash		94cbd4f7d63d0bde686ba2f3b4d8ed4eabe
Stored verification hash		94cbd4f7d63d0bde686ba2f3b4d8ed4eabe
Report Hash		94cbd4f7d63d0bde686ba2f3b4d8ed4eabe
Verify result		Match
Bad Sector List		
Bad sector(s)		No bad sectors found

Close

Bad Sector List:

A bad sector is an unusable part or partition of a disk track located on a computer hard drive or flash drive. It is usually caused by physical damage or the inability of the OS to access information in rare cases.

Conclusions

In this lab, I learned how to acquire and transfer digital evidence using ProDiscover and FTK Imager while preserving data integrity. ProDiscover creates images in the EVE file format and requires manual hashing after acquisition, whereas FTK Imager creates E01 images and automatically generates hash values during the imaging process.

Additionally, FTK Imager produces a test file containing examiner information, case details, and verification hashes. Through this lab, I gained a deeper understanding of forensic imaging, hashing algorithms such as MD5 and SHA-1, and the differences between EVE and E01 file formats.

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